PampelMuse

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Abstract

PampelMuse is a software package with the purpose of facilitating the analysis of integral-field spectroscopic observations of crowded stellar fields. It provides several subroutines to perform the individual steps of the data analysis.

1 Getting started

1.1 Requirements

PampelMuse is written in *python*. A recent version of *python* should be available on the system, *python v2.7* is recommended. In addition, several *python* packages are required to run Pampelmuse:

- *numpy*[1], tested for version 1.6.1
- *scipy*[2], tested for versions ≥0.9, a version ≥0.10 is recommended because a faster algorithm for sparse matrix inversion was introduced in v0.10
- *astropy*[3], tested for version 0.2.5

Note that the versions of the packages listed above are just a recommendation. The code may also run with earlier versions. In addition to these packages, it is highly recommended that the following packages are available on the system:

- *matplotlib*[4], tested for version 1.1.1
- *PyQt4*[5]

These packages are required in order to work with the graphical user interface (GUI) that is provided with PampelMuse . When starting PampelMuse the availability of *matplotlib* and *PyQt4* is checked. If the packages are not available, the subroutines that implement the GUI will be disabled. Finally, the packages listed below are not required to run the core packages of PampelMuse , but some auxiliary routines depend on them:

- *astLib*[6], tested for version 0.7.0
- *pyraf*[7]

1.2 Download and Installation

There are currently two possibilities to obtain PampelMuse that will be described in the following.

[1] http://www.numpy.org/
[2] http://scipy.org/
[3] http://www.astropy.org/
[4] http://matplotlib.org/
[5] http://www.riverbankcomputing.co.uk/software/pyqt/intro
[6] http://astlib.sourceforge.net/
[7] http://www.stsci.edu/institute/software_hardware/pyraf

1.2.1 Using the tarball

A zipped version of the code is available at `http://path.to/pampelmuse.tar.gz`. After the download is completed, change to the folder containing the tarball and execute the following commands

```
# tar -xvzf pampelmuse.tar.gz
# cd pampelmuse/
# python setup.py install
```

This will install PampelMuse to your default *python* path. Note that the last step usually requires root privileges. If you do not have those privileges, you can define a custom path using the `--prefix` or `--home` arguments. You only have to make sure then that the path you are using to install PampelMuse is included in your `PYTHONPATH`. For example, the following commands will install PampelMuse to folders `$HOME/lib` and `$HOME/bin`.

```
# python setup.py install --home=~
# export PYTHONPATH=$PYTHONPATH:$HOME/lib/python ## for a bash-shell or
# setenv PYTHONPATH $PYTHONPATH:$HOME/lib/python ## for a c-shell
```

You probably want to store the modification of your `PYTHONPATH` variable in your shell configuration file (`.bashrc` or `.cshrc`), otherwise you have to set the variable again every time you open a new shell. For further informations on custom installation paths, check the *python* documentation.[8]

Afterwards, you should be able to execute

```
# PampelMuse --help
```

If no error message is returned, you have installed PampelMuse successfully!

1.3 Using the SVN repository

The maintenance of PampelMuse is handled via an SVN repository located at `https://subversion.assembla.com/svn/kamann-code.2`. The advantage of using the repository is that it allows you to always use the most recent version of the code. However, the disadvantage is that when updating the code I sometimes commit revisions that are not fully operational. So when updating to a new revision, be sure to check the most recent log entries.

To install from the SVN repository, first cd into the folder were you want to install the code and check out the repository:

```
# svn checkout https://subversion.assembla.com/svn/kamann-code.2/trunk
```

This should download the most recent version of PampelMuse . To use the code afterwards, the following modifications should be added to your shell configuration:

```
# ## for a bash shell:
# export pampelmuse_path=<path to PampelMuse> ## folder were SVN checkout was performed
# export PYTHONPATH=$PYTHONPATH:$pampelmuse_path
# export PATH=$PATH:$pampelmuse_path/pampelmuse/bin
```

Again, calling

```
# PampelMuse --help
```

should return no error message.

2 The input data

2.1 Preparing the data

First note that PampelMuse is *not* a data reduction pipeline. All analysis steps assume that the IFS data has been properly reduced and that all the instrumental artefacts have been removed. In general, the IFS data can come in two flavours:

1. As a three-dimensional data cube with two spatial axes (`NAXIS1` and `NAXIS2`) and a spectral axis (`NAXIS3`). This format can always be used, even if the data were obtained with an instrument that is unknown to PampelMuse .

[8] `http://docs.python.org/2/install/#alternate-installation`

Type	required?	EXTNAME
Data	y	DATA
Uncertainties	(y)	ERROR, SIGMA
Variances	(y)	STAT
Bad pixel mask	n	MASK

Table 1: Extension names used to identify the input required for the analysis in a combined FITS file. Note that either uncertainties or variances must be provided.

2. As a two-dimensional "image" with one spatial and one spectral axis. The advantage of this format is that it can also be used with instruments that have a ireegular (i.e. non-rectangular) spatial sampling. However, for this format to work PampelMuse must know how to arrange to different spaxels in the field of view. Currently this format is only supported for PMAS and ARGUS data.

Besides the actual data, PampelMuse also requires an estimate of the uncertainties or the variances for the analysis. Optionally, a bad pixel mask can be used. Data, uncertainties, and mask can either be provided as different extensions of a single FITS file or as three individual FITS files. In the former case, the routine will use the extension names to identify what is stored in which extension, see Table 1. In the latter case, the individual filenames should have the same prefix and be suffixed `.dat.fits` (the data), `.err.fits` (the uncertainties), and `.msk.fits` (the bad pixel mask).

If provided, the bad pixel mask should be zero for valid pixels and one for pixels that should be excluded. Instead of providing a dedicated mask, one can also set the invalid pixels to NaN and PampelMuse will ignore them in the analysis.

2.2 The reference catalogue

As PampelMuse does not currently include a dedicated source detection algorithm, the analysis relies on the presence of a reference catalogue, containing coordinates and magnitudes of the stars in and around the observed field. The best sources for such a catalogue are imaging data obtained with the *Hubble* space telescope (HST) or ground based adaptive optics images. Note that the quality of the reference catalogue (especially its completeness and astrometric accuracy) will strongly influence the results of the analysis. For example, a common shortcoming of catalogues generated from HST data is that the brightest stars are often missing because they were saturated in the exposures. If this is the case, try to add them manually.

The reference catalogue should be present as a plain ascii-file with the first (non-commented) line containing the column names. PampelMuse will recognize column names such as `ra`, `dec`, or `id` and process the column content accordingly. The reference catalogue must contain the following columns:

1. A unique identifer for every source; must be a non-negative integer
2. Two columns containing the source coordinates in a reference system. Both pixel and world coordinates are accepted, and priority is given to the latter if both are present. If provided, the world coordinates should be in units of decimal degrees.
3. A magnitude for every source in a passbband that covers at least some part of the spectral range of the IFS data.

Here is an example of how a valid reference catalogue file may look like:

```
## Example reference catalogue
## Random Cluster NGC9999, generated from HST/WFPC2 data
id    ra    dec   f606w
 1  12.764311  -26.986152  13.29
 2  12.751325  -26.986431  14.87
 4  12.760123  -26.976514  12.25
 ...
```

TBD: Add possibility to include filtercurve if it is missing.

Name	Type	Default	Choices	Description
PSFTYPE	string	moffat	moffat, gaussian	The type of analytical PSF profile to be used.
PSFPARS	tuple	(2.5, 3.0, 0.0, 0.0)	–	The initial guesses for the PSF parameters. For a Moffat profile, the order is (β, FWHM, e, θ). For a Gaussian profile, the order is (FWHM, e, θ).
PSFFIT	integer	6	≥ 0	The PSF behaviour in the fit. PSFFIT = $\sum_n 2^n$, where $n \geq 1$ runs over the indices of the free parameters; e.g. PSFFIT = 6 for a Moffat means to fit β and FWHM at constant e and θ.
PSFRAD	float	10	> 0	The centroid distance out to which the profile will be calculated, given in spaxels.

Table 2: Configuration parameters influencing the PSF

3 The PSF models

PampelMuse currently uses purely analytical PSF profiles. The user can choose between a Gaussian and a Moffat profile. An extensive discussion of the profiles and all associated formulae can be found in Appendix B. Here we restrict the discussion to the main formulae and most important properties. The analytical profile of the Gaussian is

$$G(r) = \Sigma_0 \exp\left\{\frac{-r^2}{2\sigma^2}\right\}, \tag{1}$$

whereas the Moffat profile is described via

$$M(r) = \Sigma_0 \left[1 + \left(\frac{r}{r_\mathrm{d}}\right)^2\right]^{-\beta}. \tag{2}$$

In both cases, Σ_0 is the central intensity. The width of the Gaussian is controlled by the standard deviation σ, while the effective radius r_d defines the width of the Moffat. However, instead of σ and β, PampelMuse uses the FWHM as a measure for the profile width (see Appendix B).

The Moffat profile has a second shape parameter, β. It measures the kurtosis of the profile, i.e. the strength of the wings of the PSF. Note that higher values of β correspond to less pronounced wings. From my experience, $\beta \sim 2.5$ is a typical value for a seeing-limited PSF. For this value, a Moffat profile has much more pronounced wings than a Gaussian profile. Therefore I strongly recommend using the Moffat profile because the Gaussian profile will almost certainly underestimate the wings of the PSF.

The radius r incorporates the ellipticity e and the position angle θ of each profile. Thus, besides the actual centroid coordinates, the Moffat profile has 4 free parameters, the Gaussian profile has 3.

The properties of the PSF in the analysis can be controlled using a number of configuration parameters, summarized in Table 2.

4 The analysis with PampelMuse

PampelMuse provides three main routines that perform the source extraction: INITFIT, CUBEFIT, and POLYFIT. They are described in the following. There are also several auxiliary routines available for various other tasks. To get an overview of all the available routines, type

```
# PampelMuse -h
```

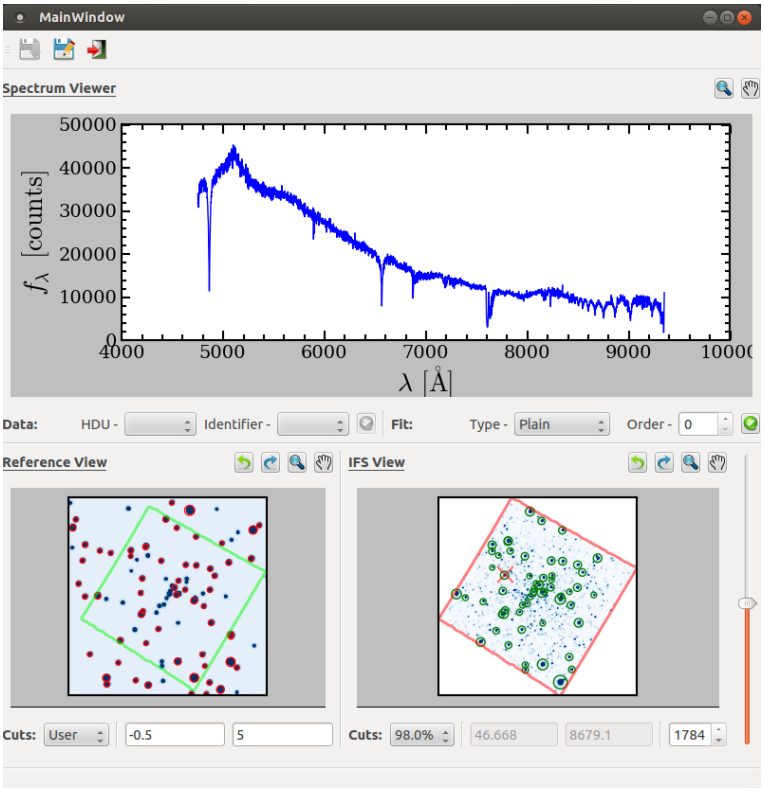


Figure 1: The graphical user interface (GUI) of PampelMuse . The bottom right panel shows a MUSE datacube of NGC 6397 observed during commissioning. The spectrum of the spaxel marked by a red cross is shown in the upper panel. The lower left panel shows a mock image created from a photometric catalogue.

4.1 The PampelMuse Gui

Before describing the individual routines of PampelMuse , the graphical user interface (GUI) that is part of several routines is introduced. The GUI is loaded automatically when running PampelMuse in interactive mode and executing the INITFIT (Sect. 4.2) or the POLYFIT (Sect. **??**) routine. An example of how the GUI looks like is displayed in Fig. 1.

The GUI shows three separate panels. The upper half is reserved for the spectrum viewer while the lower half contains the reference viewer on the left and the IFS viewer on the right side. The buttons to the top right of each panel allow to zoom in to and out of the data that is displayed, pan over it, and recover previous views of it. The cuts used to display the reference and the IFS data can be changed using the buttons at the bottom of either panel. Finally, the slider to the right of the IFS viewer can be used to scroll through the different layers of the IFS data.

4.2 INITFIT: Selecting the sources

The first step of the analysis is to identify those sources from the reference catalogue for which it is feasible to extract spectra from the IFS data. As the reference catalogue ideally was obtained from higher resolution data and typically covers a larger area on the sky, it would not make much sense to try and extract a spectrum for each star in the catalogue. Instead, we use INITFIT to make a selection. The syntax for calling this routine is

```
# PampelMuse <config> <IFS data prefix> INITFIT <reference catalogue> .
```

Note that the IFS data should be provided without the `.fits` or `.dat.fits` suffix. `config` refers to the user setup. This is explained in detail in Sect. A.

INITFIT performs the following steps in order to select a useful subset of the available sources.

1. Opening the IFS data, checking their validity.

2. Opening the reference catalogue file, checking its validity.

3. Creating an image from the IFS data in the passband in which the magnitudes in the reference catalogue were provided.

4. Creating a mock image from the sources in the reference catalogue, again in the passband in which the magnitudes were provided. The mock image is prepared using the same pixelsize as the IFS data and the initial guess of the PSF for the IFS data (see parameters `PSFTYPE` and `PSFPARS`, Sect. A).
   Note: Depending on the coverage of the reference catalogue and on the number of sources in it, this step can take a lot of time and/or consume a lot of memory. If you encounter one of these problems, you can either manually reduce the footprint of the catalogue or set the `RESLIM` parameter to a smaller magnitude. The latter works because only sources brighter than the assumed confusion limit are used to create the mock image. If at the same time the parameter `UPDATERL` is true (recommended), then this will have no effect on the final source selection.

5. Cross-correlating the two images just created to find the most likely location of the IFS field of view with respect to the reference catalogue. Note that this step requires that both images indeed have the same pixel scale and orientation. To ensure this, RA and Dec coordinates of the sources must be provided with the reference catalogue and the header of the IFS data must include a minimum of WCS information: RA and Dec of the pointing, spatial sampling and rotation angle. Otherwise this step is skipped.

6. Loading the PampelMuse GUI with the content following content (also in Fig. 1):

   - The *Reference view* shows the mock image created from the photometric input catalogue. The sources are marked with red circles and are pickable. When selecting a source via the left mouse button, the GUI displays some information about it in the status bar at the bottom of the GUI. If the cross correlation in the previous step was successful, then the reference view also shows the suggested position and orientation of the IFS field of view.
   - The *IFS view* shows a layer from datacube (initially the first one). If the cross-correlation in the previous step was successful, then the GUI will show the suggested positions of the brighter sources as green circles. It is possible to grab individual spaxels from the cube using the left mouse button.
   - The *Spectrum viewer* is initially empty. If a spaxel is grabbed in the IFS view, then the spectrum of that spaxel is displayed.

7. From here, there are basically two options on how to proceed:

   (a) If the cross correlation was successful and you are happy with the location of the IFS field of view with respect to the reference catalogue, then you can just close the GUI and let INITFIT proceed with the source selection.

   (b) If the cross correlation failed or you are not happy with its result, then you will have to define the approximate location of the IFS field of view with respect to the reference catalogue manually. To do so, proceed as follows. Pick a source in the reference catalogue, then select its approximate position in the datacube. Once a source is successfully identified, it will be displayed by a filled red circle in the Reference view. To undo the identification for a source, pick it with the *right* mouse button in the Reference view. At least three sources should be identified before closing the GUI. INITFIT will then use this information to re-determine the location of the IFS field of view.

8. Knowing the positions of the sources in the IFS field of view, INITFIT will then estimate the signal-to-noise ratio (S/N) for all sources from the reference catalogue. Depending on the size of the reference catalogue, this step may take some time, even though it is parallelized.

9. The next step is the actual source selection. INITFIT will loop over the sources in the reference catalogue – sorted by magnitude, i.e. brightest first – and do a number of checks to determine the status of every source in the fitting process:

- Check if source is brighter than the fixed confusion limit `RESLIM`. This is only done if `UPDATERL` is False (not recommended).
- Check if the local source density is below the provided threshold, `CONFDENS`. As for a larger field of view, such as the MUSE WFM, the stellar density can be very different across its extent, the confusion limit is estimated locally in an area of radius $10 \times$ FWHM around each star.
- Check if S/N is above the threshold value, `SNRCUT`. Note that the S/N estimated in the previous step is modified in two cases.
  (a) If a source is outside the IFS field of view, the S/N is penalized using a factor $\exp(-(d/\mathrm{FWHM})^2)$, where d is the distance to the nearest spaxel.
  (b) If a source is nearby a brighter source (with a distance $r <$ FWHM), the S/N is modified with the factor $1. + 0.9\log(r/\mathrm{FWHM})$. This factor approximates the decrease in S/N found for nearby sources in Kamann et al. (2013).

Sources that passed all the above checks are assigned status 2. They are considered resolved and their spectra are set for extraction. Sources that fail the S/N criterion because of the vicinity of a brighter source are given status 1 and their contribution is accounted for in the extraction of the spectrum of the brighter neighbour. All other stars are given status -1 and are ignored during the spectrum extraction.

10. The next step is only carried out if an unresolved stellar component should be included in the extraction process. This is determined by the parameters `USEUNRES` and `UNRESBIN`. If `USEUNRES` is True, the routine will collect set all sources inside the IFS field of view that are less than `UNRESBIN` magnitudes fainter than the faintest resolved source, i.e. the faintest source with status 2, to status 0 and extract a combined spectrum for them.

11. Whether or not the next step is carried out depends on the value of the parameter `SMALLIFU`. If it is set to False, then INITFIT will search for suitable sources to model the PSF during the extraction. Potential PSF sources are those that fulfil the following criteria:

    - at least 3 magnitudes brighter than provided resolution limit
    - identified as resolved (i.e. status 2) in the source selection
    - an estimated S/N > 20
    - a distance $\geq$ FWHM from the edge of the IFS field of view
    - two isolation criteria:
      - The relative contribution from neighbouring sources inside the region where the PSF is modelled (`PSFRAD`) must be smaller than the value of parameter `APERNOIS`.
      - No brighter sources are allowed inside $1.5\times$ `PSFRAD` distance.

    Aperture photometry is performed on each layer for each source that fulfils the above criteria. After that, the resulting spectra are used to redetermine the S/N estimate of each source and to perform the final selection of PSF sources. Also, the centroids of the aperture photometry are used to estimate the source coordinates as a function of wavelength. Finally, the second order moments measured for the PSF sources are used to estimate the ellipticity, and the position angle of the PSF (if they are free parameters in the PSF fit).

12. Write the results from the source selection to a dedicated multi-extension FITS-file, named *<prefix>.prm.fits* and hereafter called PRM-file. This file is loaded by all other routines of PampelMuse . An overview on the structure of the PRM-file is given in Table 3.

EXTNAME	Type	Purpose
PRIMARY	Primary	Header contains the configuration of PampelMuse
SOURCES	Binary Table	The source catalogue, contains basic information about every source used in the extraction.
SPECTRA	Image	Contains the extracted spectra and an estimate of the uncertainties.
POSPARS	Binary Table	Contains the coordinate transformation from reference to IFS coordinates.
POSITIONS	Binary Table	Contains the parameters of the polynomial fits to the IFS coordinates.
PSFPARS	Binary Table	Contains the parameters of the fitted PSF

Table 3: The different extensions of the PRM-file and their respective contents.

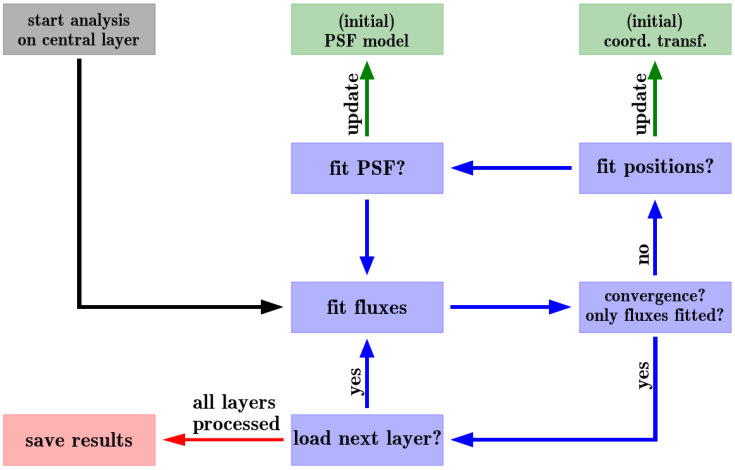


Figure 2: The iterative fittting approach of the CUBEFIT routine.

4.3 CUBEFIT - Extracting the spectra

CUBEFIT is the routine that actually performs the extraction of the spectra by simultaneously fitting a PSF to all sources that have been identified as resolvable by INITFIT. The calling sequence for CUBEFIT is

```
# PampelMuse <IFS data prefix> CUBEFIT .
```

Note that the routine expects a file PRM-file generated by INITFIT. Further note that currently it has no effect to provide a configuration to CUBEFIT. The routine will use the one that is available in the header of the PRM-file. If you find that any parameter should be changed, you will have to modify the primary header of this file.

CUBEFIT analyses the IFS data layer by layer, starting from the central layer and simulateously proceeding into the red and the blue part of the cube. On each layer, the code performs an interative analysis in order to extract the optimal flux for each source. This iterative scheme, also depicted in Fig. 2, can be summarized as follows.

1. Using the initial guesses for the PSF and the coordinate transformation, a first fit of the fluxes is performed. The flux fit is a linear least-squares fit that can be solved by direct matrix inversion and does not require initial guesses. The matrix contains the PSF profiles of the individual components. As each PSF profile is significantly different from zero only in the vicinity of the represented star, the majority of the matrix entries are zero. For such sparse matrices very efficient inversion algorithms are available. CUBEFIT uses the method `lsmr` (Fong and Saunders, 2010) from the `scipy.sparse.linalg` package. It was tested for up to 10 000 sources and provided reliable solutions in reasonable computation times.

2. With an initial value for all fluxes available, CUBEFIT updates the coordinate transformation

from reference to IFS coordinates via the Levenberg-Marquardt algorithm `leastsq` from the `scipy.optimize` package. To speed up the computation, CUBEFIT internally calculates the Jacobi matrix, i.e. the matrix containing the partial derivatives of How this is done depends on the value of the parameter `SMALLIFU`.

3. The PSF fit uses the fluxes and the updated coordinate transformation.

References

D. Fong and M. Saunders. LSMR: An iterative algorithm for sparse least-squares problems. *ArXiv e-prints*, June 2010.

S. Kamann, L. Wisotzki, and M. M. Roth. Resolving stellar populations with crowded field 3D spectroscopy. *A&A*, 549:A71, January 2013. doi: 10.1051/0004-6361/201220476.

A The configuration parameters

```
<?xml version="1.0" ?><?xml-stylesheet type="text/xsl" href="PMstyle.xsl"?>
    <ns0:configuration timestamp="2014-07-10T08:59:24Z" xmlns:ns0="http://
    www.tobedefined.com/pampelmuse" xmlns:xsi="http://www.w3.org/2001/
    XMLSchema-instance" xsi:schemaLocation="http://www.tobedefined.com/
    pampelmuse PMschema.xsd">

 <ns0:parameter category="1pho" inuse="0">
  <ns0:name>CFPCOLS</ns0:name>
  <ns0:value dtype="tuple">(0, 1, 2, 3)</ns0:value>
  <ns0:comment>The columns of the crowded-field photometry file that
      contain: ID, x position, y position and magnitude of the stars</
      ns0:comment>
  <ns0:updated timestamp="2011-08-09T16:23:06Z">Appended parameter CFPCOLS
      to configuration file.</ns0:updated>
 <ns0:updated timestamp="2013-07-25T09:45:11Z">Disabled usage of parameter.
    </ns0:updated></ns0:parameter>
 <ns0:parameter category="1pho" inuse="1">
   <ns0:name>PASSBAND</ns0:name>
   <ns0:value dtype="str">F814W</ns0:value>
   <ns0:comment>The photometric passband in which the provided magnitudes
       are given</ns0:comment>
   <ns0:updated timestamp="2011-11-29T21:03:10Z">Appended parameter
       PASSBAND to configuration file.</ns0:updated>
  </ns0:parameter>
  <ns0:parameter category="1pho" inuse="1"><ns0:name>ZEROMAG</ns0:name><
      ns0:value dtype="float">99.9</ns0:value><ns0:comment>The magnitude
      zeropoint of the IFS data in the provided passband, i.e. the magnitude
       of a source triggering 1 count/layer. Set this parameter to 99.9 to
      let the routine determine the zeropoint from the data.</ns0:comment><
      ns0:updated timestamp="2013-06-10T20:45:38Z">Appended parameter
      ZEROMAG to configuration file.</ns0:updated></ns0:parameter><
      ns0:parameter category="1pho" inuse="1">
  <ns0:name>PIXSCALE</ns0:name>
  <ns0:value dtype="float">1.0</ns0:value>
  <ns0:comment>The spaxel scale if the observation. Note that this
      parameter is used only if PampelMuse can not determine the spaxel
      scale from the FITS header of the data.</ns0:comment><ns0:updated
```

```
    timestamp="2012-03-14T09:13:10Z">Appended parameter PIXSCALE to
    configuration file.</ns0:updated>
</ns0:parameter>
<ns0:parameter category="2sel" inuse="1">
 <ns0:name>APERRAD</ns0:name>
 <ns0:value dtype="float">3.0</ns0:value>
 <ns0:comment>Radius (in spaxels) used for aperture spectrophotometry</
    ns0:comment>
 <ns0:comment>Note that this parameter only has an effect if the SMALLIFU
    flag is unset.</ns0:comment><ns0:updated timestamp="2011-08-09
    T16:26:44Z">Appended parameter APERRAD to configuration file.</
    ns0:updated>
</ns0:parameter>
<ns0:parameter category="2sel" inuse="1">
 <ns0:name>APERNOIS</ns0:name>
 <ns0:value dtype="float">0.1</ns0:value>
 <ns0:comment>The maximum allowed fraction of contaminating flux within
    APERRAD. Aperture spectrophotometry is performed only for sources
    whose nearby neighbors contribute less than APERNOIS to the measured
    flux.</ns0:comment>
 <ns0:updated timestamp="2011-08-31T08:10:40Z">Appended parameter APERNOIS
     to configuration file.</ns0:updated>
</ns0:parameter>
<ns0:parameter category="2sel" inuse="1">
 <ns0:name>RESLIM</ns0:name>
 <ns0:value dtype="float">16.0</ns0:value>
 <ns0:comment>The magnitude down to which stars can possibly be resolved
    in the datacube.</ns0:comment>
 <ns0:updated timestamp="2011-08-31T08:13:37Z">Appended parameter RESLIM
    to configuration file.</ns0:updated>
</ns0:parameter><ns0:parameter category="2sel" inuse="1"><ns0:name>
   UPDATERL</ns0:name><ns0:value dtype="bool">True</ns0:value><ns0:comment
   >If UPDATERL-flag is set, PampelMuse will determine the confusion limit
    and update RESLIM accordingly.</ns0:comment><ns0:updated timestamp="
   2012-04-16T16:49:15Z">Appended parameter UPDATERL to configuration file
   .</ns0:updated></ns0:parameter><ns0:parameter category="2sel" inuse="1"
   ><ns0:name>CONFDENS</ns0:name><ns0:value dtype="float">0.4</ns0:value><
   ns0:comment>The confusion limit, i.e the number of stars per resultion
   element. A resolution element is defined as the FWHM \[pixel\] of the
   PSF squared. Only used if UPDATERL-flag is set.</ns0:comment><
   ns0:updated timestamp="2012-04-16T16:52:59Z">Appended parameter
   CONFDENS to configuration file.</ns0:updated></ns0:parameter>
<ns0:parameter category="2sel" inuse="1">
 <ns0:name>SNRCUT</ns0:name>
 <ns0:value dtype="float">3.0</ns0:value>
 <ns0:comment>The cut in expected Signal-to-Noise below which no sources
    are selected. The Signal-to-Noise is determined from the estimated
    FWHM and the results of the aperture spectrophotometry .</ns0:comment>
 <ns0:updated timestamp="2011-09-01T16:18:18Z">Appended parameter SNRCUT
    to configuration file.</ns0:updated>
</ns0:parameter>
<ns0:parameter category="2sel" inuse="1">
 <ns0:name>BINRAD</ns0:name>
 <ns0:value dtype="float">0.3</ns0:value>
 <ns0:comment>The minimum spatial separation [measured in units of the
    expected FWHM] of two resolved sources. Resolved sources with
```

```
    distances smaller BINRAD are combined.</ns0:comment>
 <ns0:updated timestamp="2011-09-01T16:25:46Z">Appended parameter BINRAD
    to configuration file.</ns0:updated>
</ns0:parameter>
<ns0:parameter category="3fit" inuse="1">
 <ns0:name>GLOBSKY</ns0:name>
 <ns0:value dtype="bool">True</ns0:value>
 <ns0:comment>Flag determines whether a global sky background (i.e a
    component with constant surface brightness over the FoV) is included
    in the fit</ns0:comment>
 <ns0:updated timestamp="2011-11-28T19:40:14Z">Appended parameter GLOBSKY
    to configuration file.</ns0:updated>
</ns0:parameter>
<ns0:parameter category="2sel" inuse="1">
 <ns0:name>USEUNRES</ns0:name>
 <ns0:value dtype="bool">False</ns0:value>
 <ns0:comment>Flag indicates whether the contribution from the unresolved
    sources should be included as an additional component in the analysis.
    </ns0:comment>
 <ns0:updated timestamp="2011-09-01T16:28:29Z">Appended parameter USEUNRES
     to configuration file.</ns0:updated>
</ns0:parameter><ns0:parameter category="2sel" inuse="1"><ns0:name>
   UNRESBIN</ns0:name><ns0:value dtype="float">2.0</ns0:value><ns0:comment
   >Only has an effect if USEUNRES is enabled. In that case the
   contribution of the sources in the magnitude interval of width UNRESBIN
    below the confusion limit is combined in an unresolved stellar
   component</ns0:comment><ns0:updated timestamp="2012-04-16T16:45:23Z">
   Appended parameter UNRESBIN to configuration file.</ns0:updated></
   ns0:parameter>
<ns0:parameter category="0general" inuse="1">
 <ns0:name>SMALLIFU</ns0:name>
 <ns0:value dtype="bool">True</ns0:value>
 <ns0:comment>This flag defines how the determination of the PSF and the
    coordinate transformation is performed during the fit. If SMALLIFU is
    enabled, all sources are used for this purpose (recommended for IFUs
    with a small number of spatial pixels), otherwise bright, isolated
    stars are used (recommended for MUSE)</ns0:comment>
 <ns0:updated timestamp="2011-11-28T19:37:02Z">Appended parameter SMALLIFU
     to configuration file.</ns0:updated>
</ns0:parameter>


<ns0:parameter category="3fit" inuse="1"><ns0:name>LAYERRNG</ns0:name><
   ns0:value dtype="tuple">(0, -1)</ns0:value><ns0:comment>The spectral
   range of the datacube that should be analysed, given as the indices of
   the first and the last layer in that range. By default, the whole cube
   is analysed.</ns0:comment><ns0:updated timestamp="2013-03-26T14:10:43Z"
   >Appended parameter LAYERRNG to configuration file.</ns0:updated></
   ns0:parameter><ns0:parameter category="3fit" inuse="1">
 <ns0:name>PSFRAD</ns0:name>
 <ns0:value dtype="float">10.0</ns0:value>
 <ns0:comment>The radius out to which the PSF of a source will be defined.
     Should be large enough to avoid annuli of flux around the brighter
    sources in the residuals</ns0:comment>
 <ns0:updated timestamp="2011-10-25T13:49:50Z">Appended parameter PSFRAD
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```

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 </ns0:parameter>
 <ns0:parameter category="3fit" inuse="1">
  <ns0:name>POSFIT</ns0:name>
  <ns0:value dtype="int">2</ns0:value>
  <ns0:comment>Defines how source positions are handled in the fit:</
      ns0:comment>
  <ns0:comment>0 - Sources are fixed to their input positions.</ns0:comment
      >
  <ns0:comment>1 - Only a global offset is fitted.</ns0:comment>
  <ns0:comment>2 - A global offset and scaling are fitted.</ns0:comment>
  <ns0:comment>3 - Individual positions are fitted.</ns0:comment>
  <ns0:updated timestamp="2011-09-09T11:49:56Z">Appended parameter POSFIT
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 <ns0:parameter category="3fit" inuse="1">
  <ns0:name>PSFTYPE</ns0:name>
  <ns0:value dtype="str">moffat</ns0:value>
  <ns0:comment>The analytical profile that is used to model the PSF.</
      ns0:comment>
  <ns0:updated timestamp="2011-09-09T12:02:53Z">Appended parameter PSFTYPE
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 <ns0:parameter category="3fit" inuse="1">
  <ns0:name>PSFPARS</ns0:name>
  <ns0:value dtype="tuple">(2.5, 3.0, 0.0, 0.0)</ns0:value>
  <ns0:comment>The initial values of the PSF shape parameters. The meaning
      of the parameters for the individual profiles can be found in the
      manual.</ns0:comment>
  <ns0:updated timestamp="2011-09-09T12:08:06Z">Appended parameter PSFPARS
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 <ns0:parameter category="3fit" inuse="1">
  <ns0:name>PSFFIT</ns0:name>
  <ns0:value dtype="int">6</ns0:value>
  <ns0:comment>The degree of PSF variablility, given as SUM(2**N), with N
      being the position in the PSFPARS array (starting with 1) of a PSF
      shape parameter that should be fitted</ns0:comment>
  <ns0:updated timestamp="2011-09-09T12:11:20Z">Appended parameter PSFFIT
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 </ns0:parameter>
 <ns0:parameter inuse="1">
  <ns0:name>VERBOSE</ns0:name>
  <ns0:value dtype="int">2</ns0:value>
  <ns0:comment>The chattiness of PampelMuse, i.e. the amount of information
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  <ns0:updated timestamp="2011-11-28T20:13:30Z">Appended parameter VERBOSE
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</ns0:parameter>
<ns0:parameter category="3fit" inuse="1">
  <ns0:name>JUMPSTART</ns0:name>
  <ns0:value dtype="bool">False</ns0:value>
  <ns0:comment>Flag indicating whether the CUBEFIT routinestarts
      automatically after being called or after the START button has been
      pressed. Note that this parameter only has an effect if the USEQT flag
       is set.</ns0:comment><ns0:updated timestamp="2012-02-13T21:21:21Z">
      Appended parameter JUMPSTART to configuration file.</ns0:updated>
```

```
 </ns0:parameter>
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    ns0:value dtype="int">1</ns0:value><ns0:comment>The maximum nuber of
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    T07:38:46Z">Appended parameter MAXITER to configuration file.</
    ns0:updated></ns0:parameter>
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  <ns0:name>INTERACT</ns0:name>
  <ns0:value dtype="bool">True</ns0:value>
  <ns0:comment>The flag specifies whether the source selection and the
     polynomial fitting of the PSF parameters and source positions are
     performed interactively using the PampelMuse GUI or automatically.</
     ns0:comment>
  <ns0:updated timestamp="2014-04-28T14:13:17Z">Appended parameter INTERACT
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 <ns0:updated timestamp="2014-04-28T14:21:04Z">Changed default value of
    parameter INTERACT to True (was True)</ns0:updated></ns0:parameter>
 <ns0:parameter category="0gen" inuse="1">
  <ns0:name>USEQT</ns0:name>
  <ns0:value dtype="bool">False</ns0:value>
  <ns0:comment>Earlier versions of PampelMuse used a Qt GUI to visualize
     the progress of the CUBEFIT analysis. This feature may still be
     enabled by setting the USEQT flag. However, it is deprecated because
     it slows down the analysis and has not been maintained in a while.</
     ns0:comment><ns0:updated timestamp="2012-03-23T16:23:10Z">Appended
     parameter USEQT to configuration file.</ns0:updated>
 </ns0:parameter>
 <ns0:parameter category="0gen" inuse="1">
  <ns0:name>CPUCOUNT</ns0:name>
  <ns0:value dtype="int">1</ns0:value>
  <ns0:comment>The number of CPUs PampelMuse should use in the parallelized
     parts of the analysis. Set CPUCOUNT to -1 to use all available CPUs.<
     /ns0:comment>
  <ns0:updated timestamp="2012-03-23T16:26:22Z">Appended parameter CPUCOUNT
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<ns0:parameter category="4sub" inuse="1"><ns0:name>SUBUNRES</ns0:name><
    ns0:value dtype="bool">False</ns0:value><ns0:comment>If this flag is set
    , the contribution of the unresolved component will be subtracted when
    running the SUBTRES-script of PampelMuse.</ns0:comment><ns0:updated
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</ns0:parameter><ns0:parameter category="4sub" inuse="1"><ns0:name>SUBSKY</
    ns0:name><ns0:value dtype="bool">False</ns0:value><ns0:comment>If this
    flag is set, the contribution of the sky will be subtracted when running
     the SUBTRES-script of PampelMuse.</ns0:comment><ns0:updated timestamp="
    2012-04-17T09:41:35Z">Appended parameter SUBSKY to configuration file.</
    ns0:updated></ns0:parameter>
<ns0:parameter inuse="1"><ns0:name>IFU</ns0:name><ns0:value dtype="str">
    pmas</ns0:value><ns0:comment>The integral field unit used to observe the
     current dataset. Note that this parameter is used only if PampelMuse
    can not determine which IFU has been used from the FITS header of the
    data.</ns0:comment><ns0:updated timestamp="2012-11-09T21:44:59Z">
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    ns0:parameter>
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<ns0:parameter category="3fit" inuse="1"><ns0:name>POLYTYPE</ns0:name><ns0:value dtype="str">plain</ns0:value><ns0:comment>The type of polynomial used to fit the slopes of the source positions and PSF parameters with wavelength. Can be one of plain (normal polynomial), chebyshev, legendre, and hermit.</ns0:comment><ns0:updated timestamp="2014-06-15T15:00:51Z">Appended parameter POLYTYPE to configuration file.</ns0:updated><ns0:updated timestamp="2014-06-15T15:02:30Z">Changed default value of parameter POLYTYPE to plain (was plain)</ns0:updated><ns0:updated timestamp="2014-06-15T15:02:30Z">Changed category to 3fit (was 3fit)</ns0:updated></ns0:parameter>
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<ns0:parameter category="1pho" inuse="1"><ns0:name>CCMAXOFF</ns0:name><ns0:value dtype="float">-1.0</ns0:value><ns0:comment>The maximum allowed offset with respect to the header WCS position when locating the IFS pointing via cross correlation with the photometric catalog, given in spaxels. By default, it is set to half the size of the field of view.</ns0:comment><ns0:updated timestamp="2014-07-10T08:58:34Z">Appended parameter CCMAXOFF to configuration file.</ns0:updated></ns0:parameter></ns0:configuration>

B The PSF profiles

An overview of the functional forms of the analytical PSF profiles that PampelMuse uses. Besides the functional forms, we also provide the partial derivatives with respect to the individual profile parameters. They are used to determine the Jacobi matrix in the Levenberg-Marquard algorithm that performs the non-linear least squares fit.

B.1 Handling of positions

All profiles are defined as a function of r, with r giving the distance of a pixel (x_i, y_i) to the centroid. In the most general case, the PSF will be of elliptical shape, with its semi-major rotated by an angle θ **with respect to the y-axis**. For this reason, we define a coordinate system ($\hat{x}$, $\hat{y}$), that has the same origin as the PSF profile and has the $\hat{x}$-axis aligned with the semi-major axis. Assuming the central coordinates of profile n ($n \in [0, N)$) in the original system are (x^n, y^n), the transformation is defined as

$$\hat{x} = (x - x^n)\sin\theta - (y - y^n)\cos\theta\,, \tag{3}$$
$$\hat{y} = (x - x^n)\cos\theta + (y - y^n)\sin\theta\,. \tag{4}$$

The radius r is measured in units of pixels in direction of the semi-major axis of the profile, that is, for an arbitrary position angle θ with respect to that axis, r is calculated using

$$r(x, y) = \sqrt{\hat{x}^2 + \left(\frac{\hat{y}}{1-e}\right)^2}\,, \tag{5}$$

where e is the ellipticity of the profile.

The above equations imply that r is a function of $2N + 2$ parameters that can be free fit parameters in the PSF optimization: x^n, y^n, θ, and e. The partial derivatives of r with respect to those parameters are

$$\begin{aligned}\partial_{x^n} r \equiv \frac{\partial r}{\partial x^n} &= \left[\hat{x}^2 + \left(\frac{\hat{y}}{1-e}\right)^2\right]^{-1/2} \frac{1}{2}\left[2\hat{x}\frac{\partial \hat{x}}{\partial x^n} + \frac{2\hat{y}}{(1-e)^2}\frac{\partial \hat{y}}{\partial x^n}\right] \\ &= -\frac{1}{r}\left[\hat{x}\sin\theta + \frac{\hat{y}\cos\theta}{(1-e)^2}\right]\,, \end{aligned} \tag{6}$$

$$\begin{aligned}\partial_{y^n} r \equiv \frac{\partial r}{\partial y^n} &= \left[\hat{x}^2 + \left(\frac{\hat{y}}{1-e}\right)^2\right]^{-1/2} \frac{1}{2}\left[2\hat{x}\frac{\partial \hat{x}}{\partial y^n} + \frac{2\hat{y}}{(1-e)^2}\frac{\partial \hat{y}}{\partial y^n}\right] \\ &= \frac{1}{r}\left[\hat{x}\cos\theta - \frac{\hat{y}\sin\theta}{(1-e)^2}\right]\,, \end{aligned} \tag{7}$$

$$\begin{aligned}\partial_{\theta} r \equiv \frac{\partial r}{\partial \theta} &= \left[\hat{x}^2 + \left(\frac{\hat{y}}{1-e}\right)^2\right]^{-1/2} \frac{1}{2}\left[2\hat{x}\frac{\partial \hat{x}}{\partial \theta} + \frac{2\hat{y}}{(1-e)^2}\frac{\partial \hat{y}}{\partial \theta}\right] \\ &= \frac{\hat{x}\hat{y}}{r}\left[1 - \frac{1}{(1-e)^2}\right]\,, \end{aligned} \tag{8}$$

$$\begin{aligned}\partial_{e} r \equiv \frac{\partial r}{\partial e} &= \left[\hat{x}^2 + \left(\frac{\hat{y}}{1-e}\right)^2\right]^{-1/2} \frac{1}{2}\cdot\frac{2\hat{y}^2}{(1-e)^3}\,. \\ &= \frac{\hat{y}^2}{r\cdot(1-e)^3} \end{aligned} \tag{9}$$

However, the usual approach in PampelMuse is that we do not fit source coordinates directly, but that we assume they are already known in some reference frame and the task is to optimize the coordinate transformation from the reference system into the actual system of the datacube. Starting from the reference coordinates (u, v), we obtain the actual coordinates via

$$x^n = A\,u^n + B\,v^n + C\,, \tag{10}$$
$$y^n = A\,v^n - B\,u^n + D\,. \tag{11}$$

In this case, r is a function of 6 potentially free parameters: A, B, C, D, θ, and e. Accordingly, the partial derivatives $\partial_{x^n} r$ and $\partial_{y_n} r$ must be replaced by the partial derivatives with respect to A–D,

$$\begin{aligned}
\partial_A r \equiv \frac{\partial r}{\partial A} =& \partial_{x^n} r \cdot \frac{\partial x^n}{\partial A} + \partial_{y^n} r \cdot \frac{\partial y^n}{\partial A} = \partial_{x^n} r \cdot u^n + \partial_{y^n} r \cdot v^n \\
=& \frac{1}{r}\left[\hat{x} \cdot (v^n \cos\theta - u^n \sin\theta) - \frac{\hat{y} \cdot (u^n \cos\theta + v^n \sin\theta)}{(1-e)^2}\right], \qquad (12) \\
\partial_B r \equiv \frac{\partial r}{\partial B} =& \partial_{x^n} r \cdot \frac{\partial x^n}{\partial B} + \partial_{y^n} r \cdot \frac{\partial y^n}{\partial B} = \partial_{x^n} r \cdot v^n - \partial_{y^n} r \cdot u^n \\
=& -\frac{1}{r}\left[\hat{x} \cdot (u^n \cos\theta + v^n \sin\theta) + \frac{\hat{y} \cdot (v^n \cos\theta - u^n \sin\theta)}{(1-e)^2}\right], \qquad (13) \\
\partial_C r \equiv \frac{\partial r}{\partial C} =& \partial_{x^n} r \cdot \frac{\partial x^n}{\partial C} = \partial_{x^n} r\,, \qquad (14) \\
\partial_D r \equiv \frac{\partial r}{\partial D} =& \partial_{y^n} r \cdot \frac{\partial y^n}{\partial D} = \partial_{y^n} r\,. \qquad (15)
\end{aligned}$$

B.2 The Gaussian profile

The functional form of the Gaussian profile as it is used in PampelMuse is

$$G(r, e, \sigma, I) = \frac{I}{2\pi\sigma^2(1-e)} \exp\left\{\frac{-r^2}{2\sigma^2}\right\}. \qquad (16)$$

Note, however, that instead of the standard deviation σ, we use the FWHM to control the width of the Gaussian profile. The conversion between the two quantities is given as

$$\mathrm{FWHM} = \sqrt{8 \cdot \ln 2} \cdot \sigma\,. \qquad (17)$$

Furthermore, we control the integrated intensity I of the Gaussian (and any other) profile by providing a magnitude, i.e.,

$$I = 10^{-0.4 \cdot m}. \qquad (18)$$

The partial derivatives with respect to the individual parameters that define the shape of the profile (FWHM, σ, and θ) are calculated using the following formulae:

$$\begin{aligned}
\partial_{\mathrm{FWHM}} G &= \partial_\sigma G / \sqrt{8 \cdot \ln 2} \\
&= \exp\left\{\frac{-r^2}{2\sigma^2}\right\}\left(\frac{-I}{\pi\sigma^3(1-e)} + \frac{I \cdot r^2}{2\pi\sigma^5(1-e)}\right)/\sqrt{8 \cdot \ln 2} \\
&= G\left(\frac{r^2}{\sigma^3} - \frac{2}{\sigma}\right)/\sqrt{8 \cdot \ln 2}\,, \qquad (19) \\
\partial_e G &= \exp\left\{\frac{-r^2}{2\sigma^2}\right\}\left(\frac{I}{2\pi\sigma^2(1-e)^2} - \frac{I \cdot r \cdot \partial_e r}{2\pi\sigma^4(1-e)}\right) \\
&= G\left(\frac{1}{1-e} - \frac{r \cdot \partial_e r}{\sigma^2}\right), \qquad (20) \\
\partial_\theta G &= \frac{I}{2\pi\sigma^2(1-e)} \exp\left\{\frac{-r^2}{2\sigma^2}\right\} \cdot \frac{-r \cdot \partial_\theta r}{\sigma^2} \\
&= -G \cdot \frac{r \cdot \partial_\theta r}{\sigma^2}\,, \qquad (21)
\end{aligned}$$

where $\partial_e r$ and $\partial_\theta r$ are given as defined in Sect. B.1.

Regarding the partial derivatives with respect to any parameter connected to the centroid of the profile, the corresponding formulae given in Sect. B.1 are modified by a common factor,

$$\begin{aligned}
\partial_\nu G &= \frac{I}{2\pi\sigma^2(1-e)} \exp\left\{\frac{-r^2}{2\sigma^2}\right\} \cdot \frac{-r \cdot \partial_\nu r}{\sigma^2} \\
&= -G \cdot \frac{r \cdot \partial_\nu r}{\sigma^2}\,, \qquad (22)
\end{aligned}$$

with $\nu \in \{x^n, y^n, A, B, C, D\}$.

B.3 The Moffat profile

The functional form of the Moffat profile that is included in PampelMuse is given as

$$M(r, e, r_\mathrm{d}, \beta, I) = \frac{I \cdot (\beta - 1)}{\pi r_\mathrm{d}^2 \cdot (1 - e)} \left[1 + \left(\frac{r}{r_\mathrm{d}} \right)^2 \right]^{-\beta} , \tag{23}$$

where the width of the profile is mainly defined by the effective radius r_d and β controls the kurtosis of the profile. The integrated intensity I is the same as defined in Eq. 18. Similar to our approach for the Gaussian profile, we rather work with the FWHM instead of the effective radius r_d. The FWHM of the Moffat profile can be calculated via

$$\mathrm{FWHM} = 2\sqrt{2^{1/\beta} - 1} \cdot r_\mathrm{d} \, . \tag{24}$$

Equation 24 implies that the FWHM is a function of both, r_d and β. It should not be mixed up with the FWHM of a Gaussian profile that is a measure for the seeing during the observations. In general, the FWHM of a Moffat profile will be smaller than the seeing. This can be verified from Fig. 3, where we have plotted the FWHM ratio between a Gaussian and a Moffat profile as a function of β. The plotted ratios were obtained by fitting a Gaussian profile to the flux distribution created by a Moffat profile with given FWHM and β. Figure 3 shows that the shape of the Moffat profile approaches the shape of the Gaussian profile as β increases. Larger β values correspond to profiles with less pronounced wings. However, the Moffat profile will always have more pronounced wings than the Gaussian.

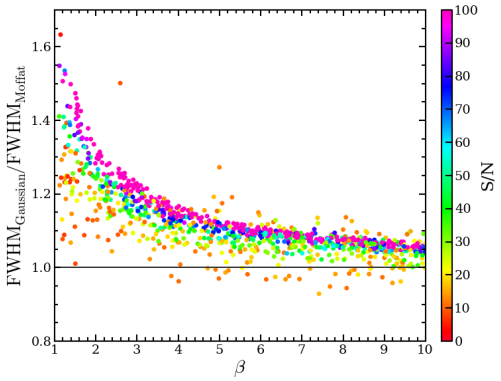


Figure 3: Inverse ratio between the FWHM of a Moffat profile and the FWHM of its best Gaussian fit, as a function of the β value of the Moffat profile and its S/N.

When calculating the partial derivatives of Eq. 23, we have to keep in mind that $r_\mathrm{d} = r_\mathrm{d}(\beta, \mathrm{FWHM})$, and therefore

$$\begin{aligned} \partial_\beta r_\mathrm{d} &= \frac{\mathrm{FWHM}}{2} \cdot \frac{-1}{(2^{1/\beta} - 1)^{-3/2}} \cdot \frac{-2^{1/\beta} \cdot \ln 2}{\beta^2} \\ &= \frac{r_\mathrm{d} \cdot \ln 2}{2 \cdot \beta^2 \cdot (1 - 2^{1/\beta})} \end{aligned} \tag{25}$$

$$\partial_\mathrm{FWHM} r_\mathrm{d} = \frac{1}{2\sqrt{2^{1/\beta} - 1}} \, . \tag{26}$$

With respect to the shape parameters FWHM, β, θ and e we thus obtain the partial derivatives

$$\begin{aligned}
\partial_{\mathrm{FWHM}} M &= \frac{-2I\cdot(\beta-1)}{\pi r_\mathrm{d}^3\cdot(1-e)}\cdot\partial_{\mathrm{FWHM}} r_\mathrm{d}\left[1+\left(\frac{r}{r_\mathrm{d}}\right)^2\right]^{-\beta} \\
&\quad - \frac{I\cdot\beta\cdot(\beta-1)}{\pi r_\mathrm{d}^2\cdot(1-e)}\left[1+\left(\frac{r}{r_\mathrm{d}}\right)^2\right]^{-\beta-1}\frac{-2r^2}{r_\mathrm{d}^3}\partial_{\mathrm{FWHM}} r_\mathrm{d} \\
&= \frac{2M}{\mathrm{FWHM}}\left[\beta\left(\frac{r}{r_\mathrm{d}}\right)^2\left[1+\left(\frac{r}{r_\mathrm{d}}\right)^2\right]^{-1} - 1\right], \qquad (27)\\
\partial_\beta M &= \left(\frac{I}{\pi r_\mathrm{d}^2\cdot(1-e)} - \frac{2I\cdot(\beta-1)}{\pi r_\mathrm{d}^3\cdot(1-e)}\cdot\partial_\beta r_\mathrm{d}\right)\left[1+\left(\frac{r}{r_\mathrm{d}}\right)^2\right]^{-\beta} \\
&\quad - \frac{I\cdot(\beta-1)}{\pi r_\mathrm{d}^2\cdot(1-e)}\left[1+\left(\frac{r}{r_\mathrm{d}}\right)^2\right]^{-\beta}\left(\ln\left[1+\left(\frac{r}{r_\mathrm{d}}\right)^2\right] - \frac{2\beta r^2}{r_\mathrm{d}^3\left[1+\left(\frac{r}{r_\mathrm{d}}\right)^2\right]}\partial_\beta r_\mathrm{d}\right) \\
&= M\left(\frac{1}{\beta-1} - \frac{2\partial_\beta r_\mathrm{d}}{r_\mathrm{d}} - \ln\left[1+\left(\frac{r}{r_\mathrm{d}}\right)^2\right] + \frac{2\beta r^2}{r_\mathrm{d}^3\left[1+\left(\frac{r}{r_\mathrm{d}}\right)^2\right]}\partial_\beta r_\mathrm{d}\right) \\
&= M\left(\frac{1}{\beta-1} - \ln\left[1+\left(\frac{r}{r_\mathrm{d}}\right)^2\right] - \frac{2\partial_\beta r_\mathrm{d}}{r_\mathrm{d}}\cdot\left[1-\frac{\beta\cdot\left(\frac{r}{r_\mathrm{d}}\right)^2}{1+\left(\frac{r}{r_\mathrm{d}}\right)^2}\right]\right), \qquad (28)\\
\partial_\theta M &= \frac{-\beta\cdot I\cdot(\beta-1)}{\pi r_\mathrm{d}^2\cdot(1-e)}\left[1+\left(\frac{r}{r_\mathrm{d}}\right)^2\right]^{-\beta-1}\frac{2r}{r_\mathrm{d}^2}\cdot\partial_\theta r \\
&= -\frac{2M\cdot\beta\cdot r}{r_\mathrm{d}^2}\left[1+\left(\frac{r}{r_\mathrm{d}}\right)^2\right]^{-1}\cdot\partial_\theta r\,, \qquad (29)\\
\partial_e M &= \frac{I\cdot(\beta-1)}{\pi r_\mathrm{d}^2\cdot(1-e)^2}\left[1+\left(\frac{r}{r_\mathrm{d}}\right)^2\right]^{-\beta} \\
&\quad - \frac{\beta\cdot I\cdot(\beta-1)}{\pi r_\mathrm{d}^2\cdot(1-e)}\left[1+\left(\frac{r}{r_\mathrm{d}}\right)^2\right]^{-\beta-1}\frac{2r}{r_\mathrm{d}^2}\cdot\partial_e r \\
&= M\left(\frac{1}{1-e} - \frac{2\beta\cdot r\cdot\partial_e r}{r_\mathrm{d}^2}\left[1+\left(\frac{r}{r_\mathrm{d}}\right)^2\right]^{-1}\right). \qquad (30)
\end{aligned}$$

Just like in the case of the Gaussian profile, the partial derivatives with respect to the parameters that define the centroid of a profile can be obtained from the formulae in Sect. B.1 via multiplication with a constant factor:

$$\begin{aligned}
\partial_\nu M &= \frac{-\beta\cdot I\cdot(\beta-1)}{\pi r_\mathrm{d}^2\cdot(1-e)}\left[1+\left(\frac{r}{r_\mathrm{d}}\right)^2\right]^{-\beta-1}\frac{2r}{r_\mathrm{d}^2}\cdot\partial_\nu r \\
&= -\frac{2M\cdot\beta\cdot r}{r_\mathrm{d}^2}\left[1+\left(\frac{r}{r_\mathrm{d}}\right)^2\right]^{-1}\cdot\partial_\nu r\,, \qquad (31)
\end{aligned}$$

where again $\nu \in \{x^n, y^n, A, B, C, D\}$.